features to light, including unusual rock strata and previously unknown fossils, whose origins were then widely debated.

The generally accepted view that Earth was only a few thousand years old had been challenged by the Comte de Buffon, and in 1785 Scottish geologist James Hutton also argued for Earth's far greater antiquity. Hutton's ideas were formed during expeditions around Scotland to examine layers of rock. He believed that Earth's crust was constantly changing, albeit mostly slowly, and could see no reason to suggest that the complex geological actions of layering, erosion, and uplifting took place faster in the distant past than they did in the present. Hutton also understood that most geological processes happen so gradually that the features he was discovering must be astronomically old.

Uniformitarianism was not generally accepted at once, not least because it challenged a literal interpretation of the creation stories of the Old Testament. However, a new generation of geologists, such as John Playfair and Charles Lyell, threw their intellectual weight behind Hutton's ideas, which also inspired a young Charles Darwin.

"… from what has actually been, we have data for concluding [what] is to happen thereafter."

James Hutton

See also: Early theories of evolution • Evolution by natural selection • Moving continents and evolution • Mass extinctions